

Hier ist die vollständige **englische Übersetzung** des bereitgestellten Dokuments „**PJG Developer Manual – IR_CIE2015 v1**“
(Strukturiert und technisch präzise, geeignet zur direkten Nutzung als Protokollreferenz.)

PJG Connection, Commands & Data Specification (IR_CIE2015 v1)

1. Serial Connection Settings

- **Baud rate:** 115200 bps
- **Parity:** None
- **Data bits:** 8
- **Stop bits:** 1
- **Checksum:** None
- **Flow control:** None

2. Data Format

All command transmissions and received data use **hexadecimal format (HEX)**.

3. Packet Format

3.1 Command Packet

Field	Header	Total Length	Command Type	Command Data	Checksum	End
Size	2 bytes	3 bytes	1 byte	Variable	1 byte	2 bytes
Value	0xCC 0x01	xx xx xx	xx	See details	Sum checksum	0x0D 0x0A

3.2 Response Packet

Field	Header	Total Length	Data Type	Payload	Checksum	End
Size	2 bytes	3 bytes	1 byte	Variable	1 byte	2 bytes
Value	0xCC 0x81	xx xx xx	xx	See details	Sum checksum	0x0D 0x0A

Notes:

- Data type in response equals the command type used to request it.
- **Checksum:** Sum of all bytes prior to checksum, keep lower 8 bits.

4. Get Spectral Start & End Wavelength

Command

- Total Length: 0x09 0x00 0x00
- Command Type: 0x0F
- Data: none

0xCC 0x01 0x09 0x00 0x00 0x0F 0xE5 0x0D 0x0A

Response

- Total Length: 0x0D 0x00 0x00
- Data Type: 0x0F
- Payload: **4 bytes**
 - Start wavelength (2 bytes, LSB first)
 - End wavelength (2 bytes, LSB first)

0xCC 0x81 0x0D 0x00 0x00 0x0F WL_START(2B) WL_END(2B) checksum 0x0D 0x0A

Example:

Spectral range = 340–1020 nm

0xCC 0x81 0x0D 0x00 0x00 0x0F 0x54 0x01 0xFC 0x03 0xBD 0x0D 0x0A

5. Acquire Single-Frame Spectrum

Command

- Total Length: 0x09 0x00 0x00
- Command Type: 0x32
- Data: none

0xCC 0x01 0x09 0x00 0x00 0x32 0x08 0x0D 0x0A

Response

Packet Length Formula (bytes):

6 + 1 + 4 + 47*4 + 3*4 + 2 + SpectrumLength*2 + 3

Segment	Type
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Segment	Type
Exposure status	uint8
Exposure time (μs)	uint32
47 photometric parameters	float[47]
3 NIR parameters	float[3]
Spectrum scale factor N	int16
Spectral data points	uint16[]

Data Definitions

1. Exposure Status (uint8)

- 0x00 = Normal
- 0x01 = Overexposed
- 0x02 = Underexposed

2. Exposure Time (uint32)

- Unit: microseconds
- Byte order: LSB first

3. Photometric Parameters (47 × float)

```
[
X, Y, Z, x, y, u, v, u', v',
CCT, Nit,
r_ratio, g_ratio, b_ratio,
DUV,
Ra,
R1, R2, R3, R4, R5, R6, R7, R8, R9, R10,
R11, R12, R13, R14, R15,
Lp, HW, Ld,
purity, SP, SDCM, k,
lux, Ee, fc,
CQS,
GAI_EES, GAI_BB_8, GAI_BB_15,
EML, M_EDI
]
```

4. Near-Infrared Parameters (3 × float)

```
[Red_Ee, Nir_EeA, Nir_EeB]
```

5. Spectrum Scale Factor (int16)

Indicates spectrum data was scaled by 10^n .

Example:

- Received value = 1300 at 340 nm
- $N = 2 \rightarrow \text{Actual value} = 1300 / 10^2 = 13$

6. Spectral Data (uint16)

- **2 bytes per value**, LSB first
- Number of samples equals wavelength range size

Example Packet

- Exposure: normal (0x00)
- Exposure time: 2500 μs
- Scale factor: $N = 2$
- Range: 340–1020 nm $\rightarrow$ **681 points**

Total length:

$6 + 1 + 4 + 47*4 + 3*4 + 2 + 681*2 + 3 = 1578 \text{ bytes}$

Packet begins:

0xCC 0x81 0x2A 0x06 0x00 0x32 0x00 0xC4 0x09 0x00 0x00 ...

6. Start Continuous Spectrum Acquisition

Command

- Type: 0x33

0xCC 0x01 0x09 0x00 0x00 0x33 0x09 0x0D 0x0A

Response

- Exactly the same data structure as single-frame acquisition.
- Frames are received continuously until a stop command is issued.

7. Stop Spectrum Acquisition

Command

- Type: 0x04

0xCC 0x01 0x09 0x00 0x00 0x04 0xDA 0x0D 0x0A

8. Get Device Information

Command

- Type: 0x08
- Data length: 0x18 (24 bytes)

0xCC 0x01 0x0A 0x00 0x00 0x08 0x18 0xF7 0x0D 0x0A

Response

- Payload: **24 ASCII bytes identifying the device**

Example:

P42B4I10234CBPD-412-0005

9. Set Exposure Mode

Command

- Type: 0x0A
- Data:
 - 0x01 = Auto exposure
 - 0x00 = Manual exposure

Example (manual mode):

0xCC 0x01 0x0A 0x00 0x00 0x0A 0x00 0xE1 0x0D 0x0A

Response:

- Success → Data 0x00
- Failure → Data 0x15

10. Get Exposure Mode

Command

- Type: 0x0B

Response data:

- 0x01 = Auto
- 0x00 = Manual

11. Set Exposure Time

Command

- Type: 0x0C
- Data: uint32 (μs, LSB first)

Example (100 ms = 100000 μs):

0xCC 0x01 ... 0x0C 0xA0 0x86 0x01 0x00 ...

Response:

- 0x00 = OK
- 0x15 = Failed

12. Get Exposure Time

Command

- Type: 0x0D

Response

- 4 bytes exposure time (μs)

13. Set Maximum Exposure Time

Command

- Type: 0x13
- Data: uint32 μs.

Example (5000 ms → 5,000,000 μs):

0xCC 0x01 ... 0x13 0x40 0x4B 0x4C 0x00 ...

14. Get Maximum Exposure Time

Command

- Type: 0x14

Payload:

- 4 bytes (μ s)

15. Set CIE Mode

Command

- Type: 0x36
- Payload:

Value	Mode
0x00	CIE 1931 – 2°
0x02	CIE 2015 – 2°
0x03	CIE 2015 – 10°

Example:

0xCC 0x01 ... 0x36 0x02 ...

Response:

- 0x00 = Success
- 0xFF = Failure

16. Get CIE Mode

Command

- Type: 0x37

Response Data:

Value	Mode
0x00	CIE1931 – 2°
0x01	CIE1964 – 10°
0x02	CIE2015 – 2°
0x03	CIE2015 – 10°

17. Send Efficiency Curve Correction Ratios

(Requires baud rate ≥ 115200)

17.1 Start Transmission Packet

Command Type: 0x23

0xCC 0x01 0x0A 0x00 0x00 0x23 0x04 ...

17.2 Ratio Data Packets

- Type: 0x23
- Payload: `float[]` correction factors
- **Maximum packet size: 999 bytes → split into multiple packets if necessary**

Example array:

`ratio[661] = {1.5, 1.5, 1.5, ...}`

`Float(1.5) = 0x3F 0xC0 0x00 0x00`

Transmission order = **LSB first**

18. Validate Ratios & Calculate Efficiency Curve

Command

- Type: 0x27

Response:

- Success → Data 0x00
- Failure → Data 0xFF

19. Factory Reset of Efficiency Curve

Command

- Type: 0x25

Response:

- 0x00 = Success
- 0xFF = Failure

Table 1 – Photometric Parameters

Parameter	Description
X, Y, Z	CIE1931 tristimulus values
x, y	CIE1931 chromaticity coordinates
u*, v*	CIE1976 chromaticity
Tc / CCT	Correlated color temperature
Nit	Luminance (nit)
r_ratio	Red ratio (%)
g_ratio	Green ratio (%)
b_ratio	Blue ratio (%)
DUV	Chromaticity deviation (
Ra	Color Rendering Index average (R1–R15)
R1–R15	Individual CRI values
Lp	Peak wavelength (nm)
HW	FWHM (half width)
Ld	Dominant wavelength (nm)
purity	Color purity (%)
S/P	Scotopic/Photopic ratio
SDCM / k	Color tolerance around CCT k
lux	Illuminance (lx)
Ee	Irradiance (W/m ²)
fc	Foot-candles
CQS	Color Quality Scale
GAI_EES	Gamut Area Index (EES spectrum)
GAI_BB_8	GAI with 8 colors, blackbody
GAI_BB_15	GAI with 15 colors, blackbody
EML	Equivalent melanopic lux
M_EDI	Melanopic EDI

Table 2 – Near Infrared Parameters

Parameter	Description	Wavelength
Red_Ee	Red irradiance (W/m ²)	701–780 nm
Nir_EeA	Near-IR irradiance A	781–800 nm
Nir_EeB	Near-IR irradiance B	> 800 nm